

Unit 4 :Chatbot Development:

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Chatbot Concepts

- A chatbot is a software application that communicates with users through text or voice.
- Chatbots use Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP).
- They are designed to simulate human conversation.
- Chatbots can answer questions, provide information, and automate tasks.

Common features:

Instant response

24/7 availability

User interaction

Automated communication

• Types of chatbots:

• Rule-based chatbots

AI-powered chatbots

Origin of Chatbot

1. ELIZA (1966) – First Chatbot

The first well-known chatbot was **ELIZA**, created in 1966 by **Joseph Weizenbaum** at the Massachusetts Institute of Technology (MIT).

ELIZA used simple pattern matching and substitution techniques.

It imitated a psychotherapist by responding with questions.

Example:

User: "I am feeling sad."

ELIZA: "Why are you feeling sad?"

Although ELIZA did not truly understand language, many users felt they were talking to a real person.

Orgin of chatbot

2.PARRY (1972)

Another important chatbot was **PARRY**, developed by psychiatrist **Kenneth Colby**.

PARRY simulated the behavior of a person with paranoid schizophrenia. It was considered more advanced than ELIZA because it modeled emotions and behavior.

3.ALICE (1995)

In 1995, **ALICE** was created by **Richard Wallace**.

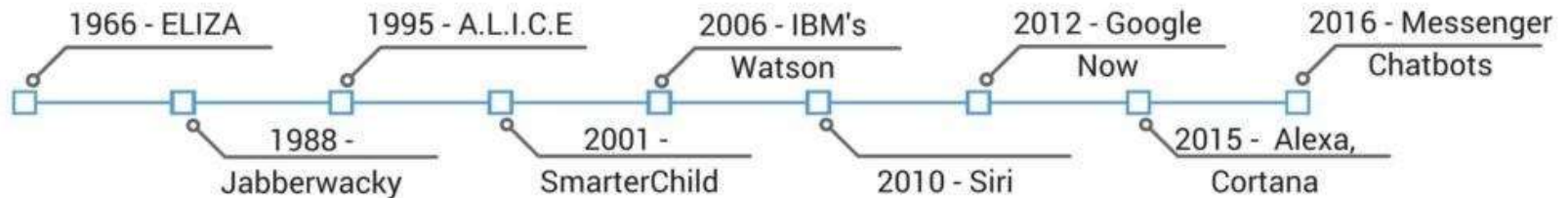
ALICE used AIML (Artificial Intelligence Markup Language).

It became one of the most influential rule-based chatbots.

Won the Loebner Prize multiple times.

Origin of chatbots

Brief History of Chatbots



Orgin of chatbots

Modern chatbots use

Deep Learning Neural Networks

Natural Language Processing (NLP)

Machine Learning

Large Language Models (LLMs)

Examples include:

ChatGPT

Google Assistant

Siri

Amazon Alexa

Evolution of chatbot

Year	Chatbot	Developer	Contribution
1966	ELIZA	Joseph Weizenbaum	First chatbot
1972	PARRY	Kenneth Colby	Simulated emotions
1995	ALICE	Richard Wallace	AIML-based chatbot
2001	SmarterChild	ActiveBuddy	Instant messaging bot
2011	Siri	Apple	Voice assistant
2014	Alexa	Amazon	Smart speaker AI
2022	ChatGPT	OpenAI	Advanced generative AI

How Does a Chatbot Work?

A **chatbot** works by receiving user input, processing the language, understanding the meaning, and generating an appropriate response. Modern chatbots use **Artificial Intelligence (AI)**, **Natural Language Processing (NLP)**, and **Machine Learning (ML)** to communicate with users.

Step 1: User Input

The user sends a message through text or voice.

Example:

“What is the weather today?”

How Does a Chatbot Work?

Step 2: Input Processing

The chatbot converts the user message into machine-readable form.

This includes:

- Tokenization
- Removing stop words
- Stemming/Lemmatization
- Syntax analysis

Example:

“What is the weather today?” → [weather, today]

How Does a Chatbot Work?

Step 3: Natural Language Understanding (NLU)

The chatbot identifies:

- **Intent** → What the user wants
- **Entities** → Important information in the sentence

Example:

- Intent: Weather inquiry
- Entity: Today.

Step 4: Response Generation

The chatbot decides how to answer.

There are two major methods:

How Does a Chatbot Work?

A. Rule-Based Chatbot

- Uses predefined rules and keywords
- Follows IF–THEN logic

Example:

IF user says "hello"

THEN reply "Hi!"

B. AI-Based Chatbot

- Uses Machine Learning and Deep Learning
- Learns from data and conversations
- Generates human-like responses

Modern AI chatbots use:

- Neural Networks
- Deep Learning
- Transformers
- Large Language Models (LLMs)

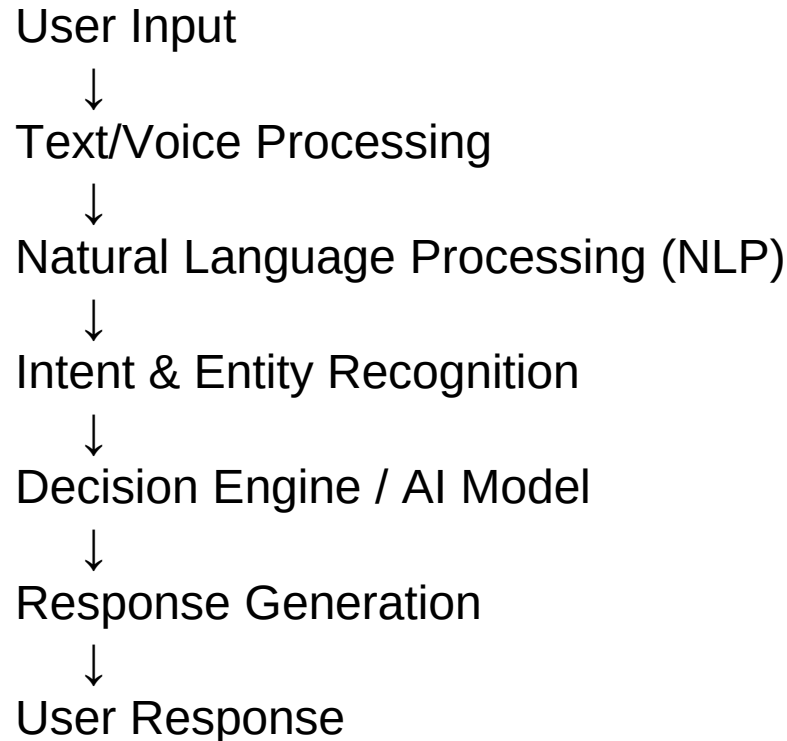
How Does a Chatbot Work?

Step 5: Response Output

The chatbot sends the final response back to the user

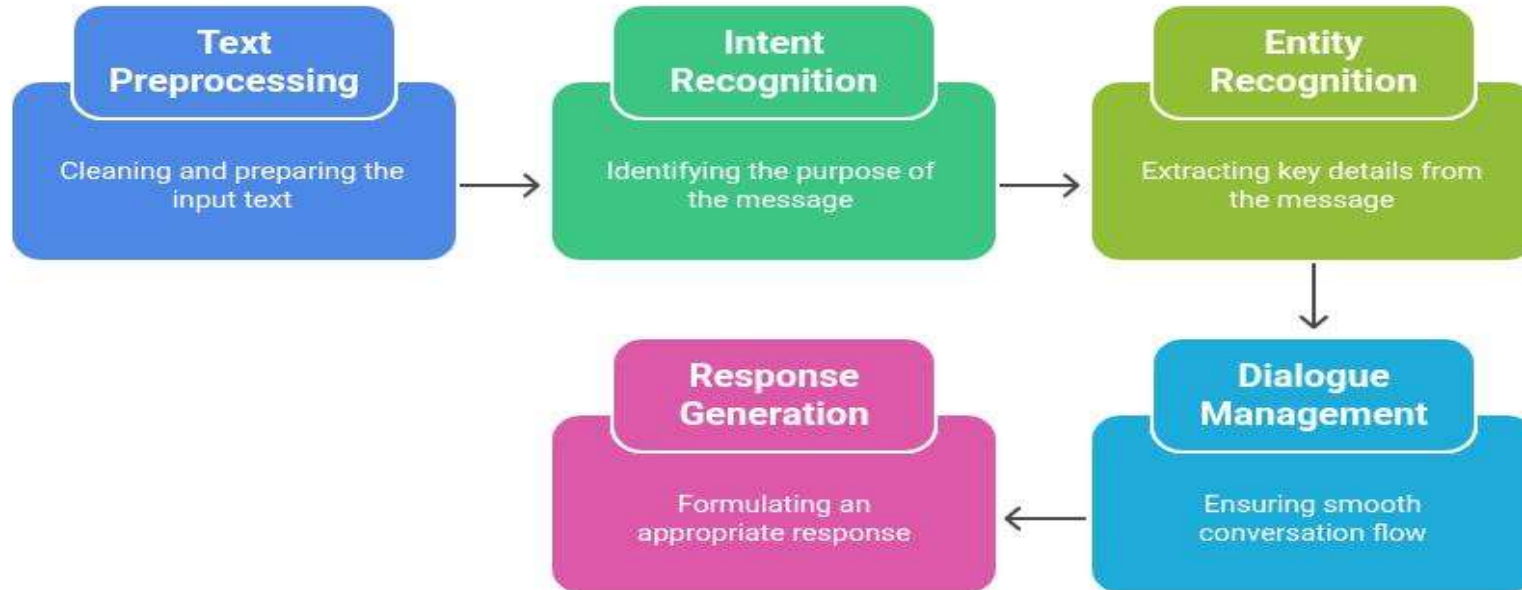
.Example:“Today’s weather is sunny.”

Chatbot Architecture



Chatbot processing Steps

Chatbot Processing Steps
Using NLP



Technologies Used in Chatbots

Technologies Used in Chatbots

Technology	Purpose
NLP	Understand human language
Machine Learning	Learn from data
Deep Learning	Improve response accuracy
Neural Networks	Process language patterns
Speech Recognition	Convert voice to text
Text-to-Speech	Convert text to voice

Why are chatbots such a big opportunity?

Chatbots are considered a major opportunity because they help organizations automate communication, improve customer experience, reduce costs, and provide services 24/7. With advances in Artificial Intelligence (AI), chatbots are becoming more intelligent and widely used across industries.

Why are chatbots such a big opportunity?

1. 24/7 Availability

Chatbots can work continuously without breaks.

Benefits:

Customer support anytime

Faster responses

Improved user satisfaction

Example:

Banking and e-commerce support bots available day and night.

Why are chatbots such a big opportunity?

2. Cost Reduction

Companies can reduce operational costs by automating repetitive tasks.
Chatbots can: Answer FAQs Handle basic customer queries Reduce the need for large support teams

3. Faster Customer Service

Chatbots provide instant replies instead of making users wait.
Advantages: Reduced waiting time Quick issue resolution
Better customer experience

4. Scalability

A chatbot can interact with thousands of users simultaneously.
Human agents:
Limited to one or few conversations at a time
Chatbots:
Handle multiple users efficiently

Why are chatbots such a big opportunity?

5. Personalized User Experience

AI-powered chatbots can analyze: User preferences Purchase history Search behavior This helps provide personalized recommendations and responses

.6. Growth of AI and NLP

Modern chatbots use: Natural Language Processing (NLP) Deep Learning Machine Learning Neural Networks These technologies improve: Language understanding Context awareness Human-like conversation

7. Business Automation

Chatbots automate many business operations such as: Appointment booking Order tracking Feedback collection Technical support This increases productivity and efficiency.

Why are chatbots such a big opportunity?

8. Multi-Platform Integration

Chatbots can work on:

- Websites
- Mobile apps
- Social media
- Messaging platforms

Examples:

WhatsApp

Telegram

Facebook Messenger

Why are chatbots such a big opportunity?

Industry	Usage
Healthcare	Patient assistance
Education	Student support
Banking	Account and transaction help
E-commerce	Shopping assistance
Travel	Ticket booking and guidance
IT Support	Technical troubleshooting

Conversational Bot

A **Conversational Bot** is an Artificial Intelligence (AI)-based software program that communicates with humans through text or voice in a natural conversational manner. It is designed to understand user queries, process language, and generate meaningful responses automatically.

Conversational bots are also called:

Chatbots

Virtual Assistants

AI Assistants

Examples include:

ChatGPT

Google Assistant

Siri

Amazon Alexa

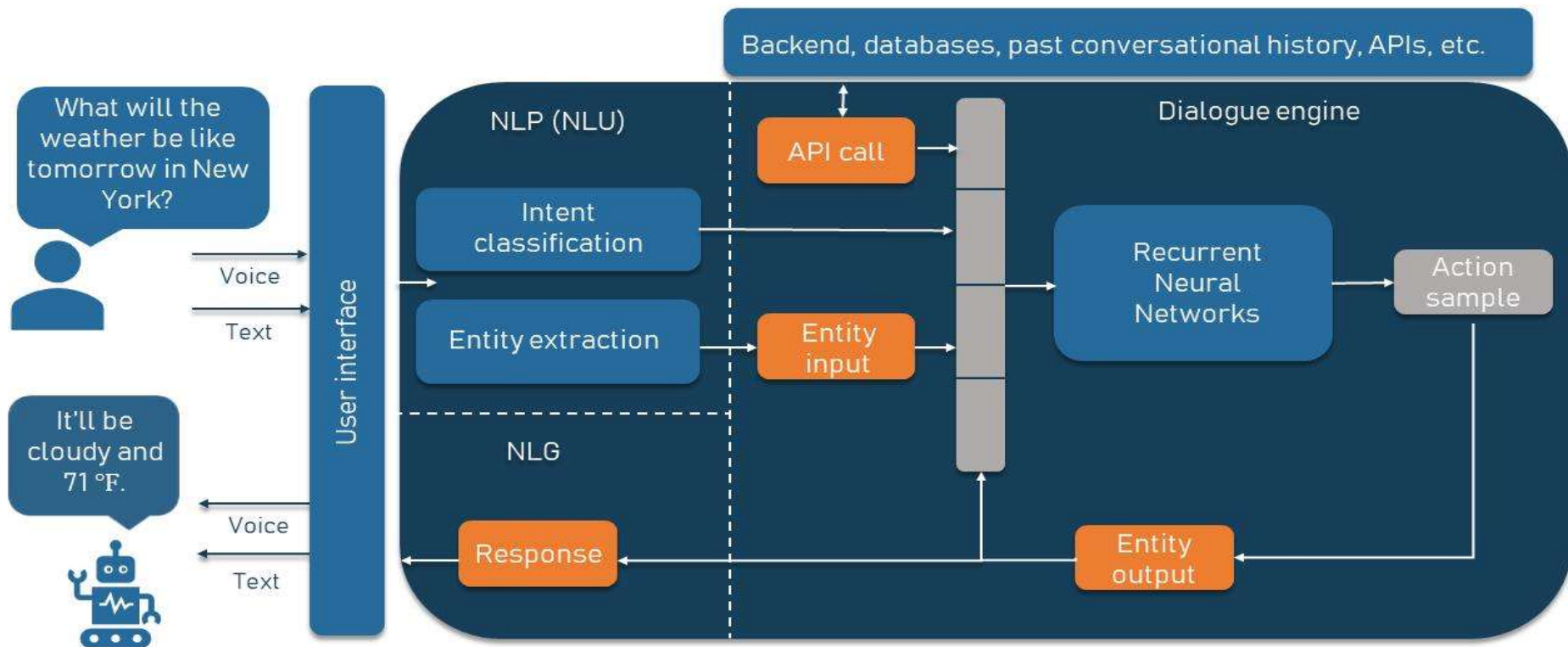
Conversational Bot

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Feature	Description
Natural Communication	Interacts like humans
24/7 Availability	Works continuously
Fast Responses	Gives instant replies
Automation	Reduces manual work
Multi-language Support	Supports different languages
Context Awareness	Understands conversation flow

Conversational Bot

HOW CONVERSATIONAL AI WORKS



Chatbot: Automatic Text Generation Procedure

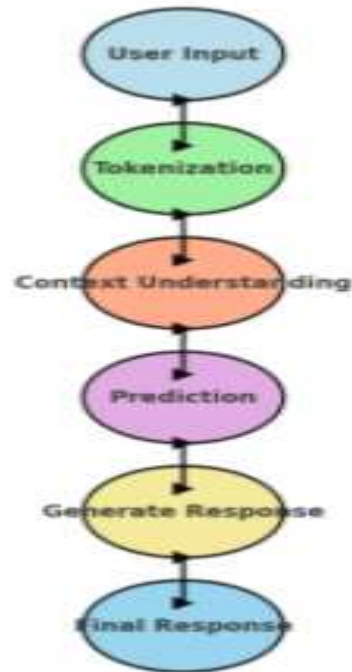
Automatic Text Generation is the process through which a chatbot automatically creates meaningful and human-like responses based on user input. Modern chatbots use Artificial Intelligence (AI), Natural Language Processing (NLP), Machine Learning (ML), and Deep Learning to generate text.

A chatbot automatically generates responses by:

- Understanding user input
- Processing language
- Predicting suitable words or sentences
- Producing meaningful output

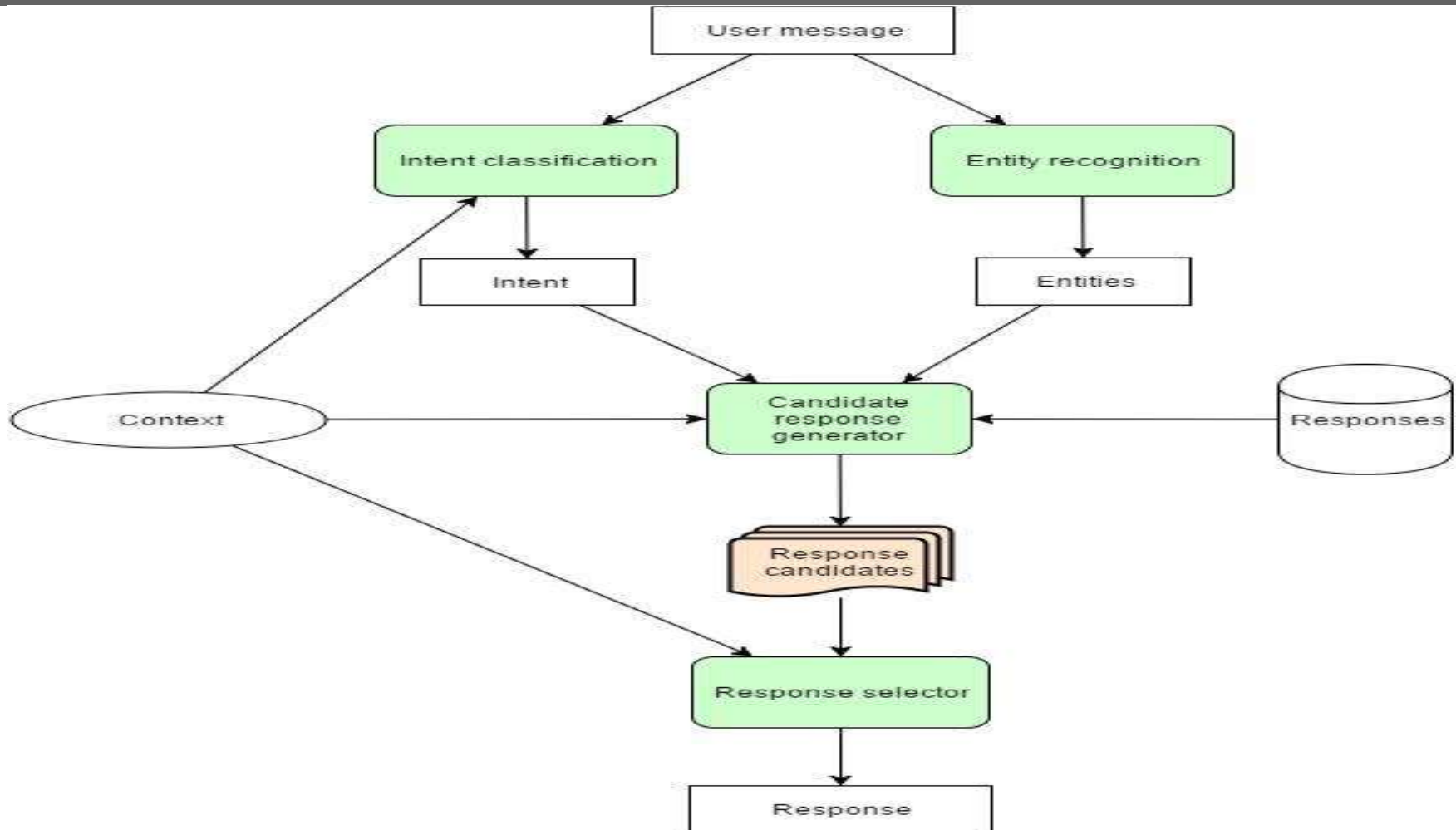
Chatbot: Automatic Text Generation Procedure

How ChatGPT Generates Responses



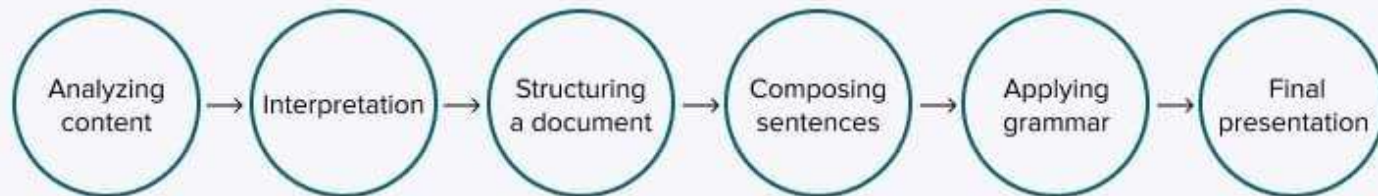
ChatGPT generates responses by understanding user input, tokenizing it, and predicting the best possible continuation. The process begins with the user input, which is broken down into tokens. ChatGPT then understands the context, and uses its learned knowledge to predict and generate the most contextually relevant response. The final response is constructed to fit naturally within the conversation.

Chatbot: Automatic Text Generation Procedure



Chatbot: Automatic Text Generation Procedure

How does natural language generation work?



Architecture of Automatic Text Generation

User Input



Text Preprocessing



Natural Language Processing



Feature Extraction



Deep Learning Model



Text Generation



Generated Response

Technologies Used

Technology	Purpose
NLP	Understand human language
Machine Learning	Learn from data
Deep Learning	Generate intelligent responses
Neural Networks	Language modeling
Transformers	Advanced text prediction

Chatbot: Automatic Text Generation Procedure

Automatic text generation in chatbots involves understanding user input, processing language using NLP, applying deep learning models, and generating meaningful responses automatically. Modern AI chatbots use advanced transformer-based architectures to provide intelligent and natural conversations.

Applications

Virtual assistants
Customer support
Language translation
Text summarization
Educational systems
Content generation

Chatbot: Automatic Text Generation Procedure

Advantages

- Fast response generation
- Human-like communication
- Supports multiple languages
- Automates customer interaction
- Learns from large datasets

Limitations

- May generate incorrect responses
- Requires large computational resources
- Difficult to understand emotions fully
- Depends on training data quality

Introduction to ChatGPT AI Chatbot

ChatGPT is an advanced Artificial Intelligence (AI) chatbot developed by [OpenAI](#). It is designed to understand human language and generate human-like responses through text-based conversation.

ChatGPT is based on:

Artificial Intelligence (AI)

Natural Language Processing (NLP)

Deep Learning

Machine Learning

Large Language Models (LLMs)

It can interact with users naturally and assist in many tasks such as answering questions, writing content, coding, tutoring, and research support.

Introduction to ChatGPT AI Chatbot

Meaning of ChatGPT

The term **ChatGPT** stands for:

Term	Meaning
Chat	Conversational interaction
GPT	Generative Pre-trained Transformer

Generative → Generates new text

Pre-trained → Trained on large datasets before use

Transformer → Deep learning architecture for language understanding

Features of ChatGPT

Feature	Description
Natural Conversation	Human-like communication
Context Understanding	Understands conversation flow
Text Generation	Creates meaningful text
Multilingual Support	Supports many languages
Question Answering	Provides detailed explanations
Coding Assistance	Helps generate and debug code
Learning Support	Assists students and researchers

How ChatGPT Works?

Step 1: User Input

The user enters a question or message.

Example: “What is Artificial Intelligence?”

Step 2: Natural Language Processing

ChatGPT processes the language using NLP techniques such as:

- Tokenization
- Context analysis
- Intent recognition

How ChatGPT Works?

Step 3: Deep Learning Model

The input is processed using Transformer-based neural networks.

Modern language models predict the next word using probability:

$$P(w_t|w_1, w_2, \dots, w_{t-1})$$

This means the model predicts the next word based on previous words in the sentence.

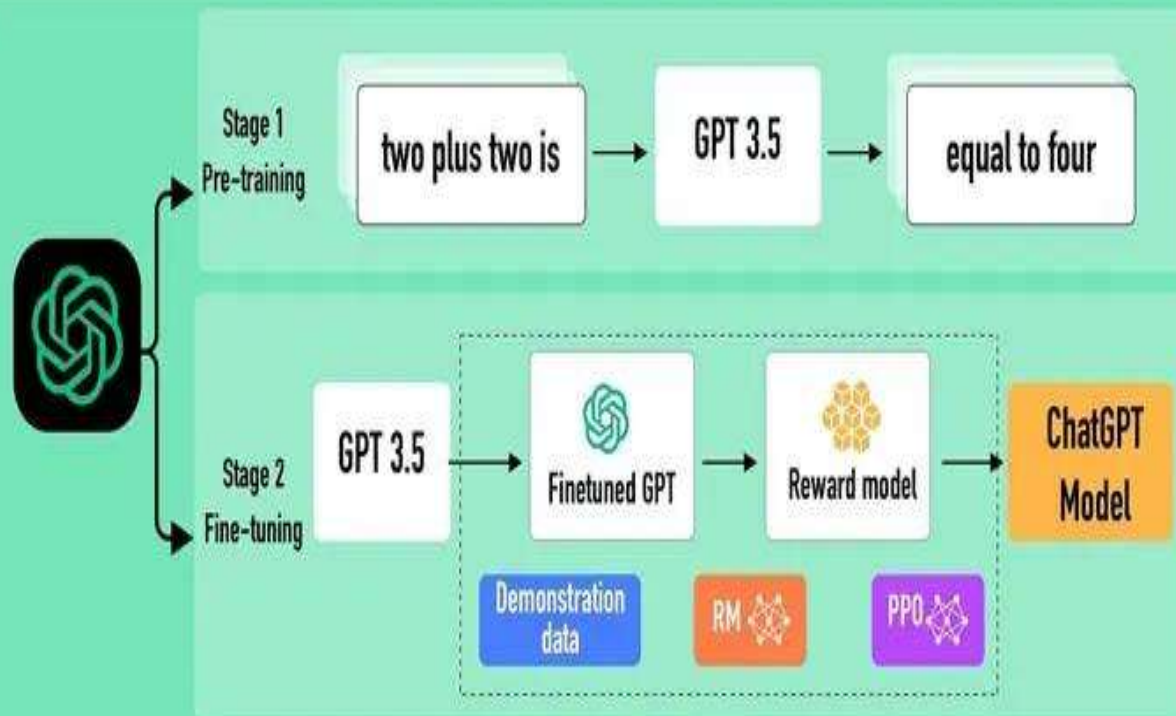
Step 4: Response Generation

The AI generates the most relevant and meaningful response. Example: "Artificial Intelligence is the simulation of human intelligence by machines."

Architecture of ChatGPT

User Input
↓
Natural Language Processing
↓
Transformer Neural Network
↓
Language Understanding
↓
Response Generation
↓
User Output

How ChatGPT Works Technically



Applications of ChatGPT

Area	Application
Education	Tutoring and explanations
Programming	Code generation and debugging
Customer Support	Automated assistance
Research	Information summarization
Content Writing	Article and report generation
Translation	Language conversion

Applications of ChatGPT

Advantages of ChatGPT

Fast response generation Human-like interaction

Handles large amounts of information

Supports multiple domains

Improves productivity Available 24/7

Limitations of ChatGPT

May generate incorrect information

Cannot fully understand emotions

Depends on training data

Requires internet and computational resources

Privacy concerns in sensitive applications

Difference Between Traditional Chatbots and ChatGPT

Traditional Chatbot	ChatGPT
Rule-based	AI-based
Limited responses	Dynamic responses
Keyword matching	Context understanding
Less flexible	Highly flexible
Predefined answers	Generates new responses

Conclusion

ChatGPT is a modern AI chatbot that uses deep learning and NLP to generate intelligent and human-like conversations. It represents a major advancement in conversational AI and is widely used in education, business, research, and automation systems.

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